

Differential Equations (2005–2014)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

20

TOTAL MARKS

103 M

TEST DURATION

60 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		20	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q16 • ID: JAM_2014_Q16)

MCQ

+2 M

Neg: -0.67

Q.16 The value of $\alpha \in \mathbb{R}$ for which the curves $x^2 + \alpha y^2 = 1$ and $y = x^2$ intersect orthogonally is

- (A) -2 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) 2

Question 2 (JAM 2014 • Q15 • ID: JAM_2014_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 The equation of the curve passing through the point $\left(\frac{\pi}{2}, 1\right)$ and having slope $\frac{\sin(x)}{x^2} - \frac{2y}{x}$ at each point (x, y) with $x \neq 0$ is

- (A) $-x^2 y + \cos(x) = \frac{-\pi^2}{4}$ (B) $x^2 y + \cos(x) = \frac{\pi^2}{4}$
(C) $x^2 y - \sin(x) = \frac{\pi^2}{4} - 1$ (D) $x^2 y + \sin(x) = \frac{\pi^2}{4} + 1$

Question 3 (JAM 2014 • Q14 • ID: JAM_2014_Q14)

MCQ

+2 M

Neg: -0.67

Q.14 Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation

$$x^2 y''(x) - 2xy'(x) - 4y(x) = 0 \text{ for } x \in [1, 10].$$

Consider the Wronskian $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. If $W(1) = 1$, then $W(3) - W(2)$ equals

- (A) 1 (B) 2 (C) 3 (D) 5

Question 4 (JAM 2014 • Q4 • ID: JAM_2014_Q4)

MCQ

+1 M

Neg: -0.33

Q.4 For $a, b, c \in \mathbb{R}$, if the differential equation

$$(ax^2 + bxy + y^2)dx + (2x^2 + cxy + y^2)dy = 0$$

is exact, then

- (A) $b = 2, c = 2a$ (B) $b = 4, c = 2$ (C) $b = 2, c = 4$ (D) $b = 2, a = 2c$

Question 5 (JAM 2012 • Q6 • ID: JAM_2012_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 An integrating factor for the differential equation $(2xy + 3x^2y + 6y^3)dx + (x^2 + 6y^2)dy = 0$ is

(A) x^3

(B) y^3

(C) e^{3x}

(D) e^{3y}

Question 6 (JAM 2012 • Q5 • ID: JAM_2012_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 The differential equation $(1 + x^2y^3 + \alpha x^2y^2)dx + (2 + x^3y^2 + x^3y)dy = 0$ is exact if α equals

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) 2

(D) 3

Question 7 (JAM 2011 • Q6 • ID: JAM_2011_Q6)

MCQ

+6 M

Neg: -2.0

Q.6 If y^a is an integrating factor of the differential equation $2xy dx - (3x^2 - y^2) dy = 0$, then the value of a is

(A) -4

(B) 4

(C) -1

(D) 1

Question 8 (JAM 2011 • Q5 • ID: JAM_2011_Q5)

MCQ

+6 M

Neg: -2.0

Q.5 The solution $y(x)$ of the differential equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$ satisfying the conditions $y(0)=4, \frac{dy}{dx}(0)=8$ is

(A) $4e^{2x}$

(B) $(16x+4)e^{-2x}$

(C) $4e^{-2x} + 16x$

(D) $4e^{-2x} + 16xe^{2x}$

Question 9 (JAM 2010 • Q10 • ID: JAM_2010_Q10)

MCQ

+6 M

Neg: -2.0

Q.10 Consider the differential equation $(x+y+1)dx + (2x+2y+1)dy = 0$. Which of the following statements is true?

- (A) The differential equation is linear
- (B) The differential equation is exact
- (C) e^{x+y} is an integrating factor of the differential equation
- (D) A suitable substitution transforms the differentiable equation to the variables separable form

Question 10 (JAM 2010 • Q9 • ID: JAM_2010_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Consider the differential equation $\frac{dy}{dx} = ay - by^2$, where $a, b > 0$ and $y(0) = y_0$. As $x \rightarrow +\infty$, the solution $y(x)$ tends to

- (A) 0
- (B) $\frac{a}{b}$
- (C) $\frac{b}{a}$
- (D) y_0

Question 11 (JAM 2009 • Q14 • ID: JAM_2009_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let $f, g: [-1, 1] \rightarrow \mathbb{R}$, $f(x) = x^3$, $g(x) = x^2|x|$. Then

- (A) f and g are linearly independent on $[-1, 1]$
- (B) f and g are linearly dependent on $[-1, 1]$
- (C) $f(x)g'(x) - f'(x)g(x)$ is NOT identically zero on $[-1, 1]$
- (D) there exist continuous functions $p(x)$ and $q(x)$ such that f and g satisfy $y'' + py' + qy = 0$ on $[-1, 1]$

Question 12 (JAM 2009 • Q12 • ID: JAM_2009_Q12)

MCQ

+6 M

Neg: -2.0

Q.12 Consider the differential equation $2\cos(y^2)dx - x\sin(y^2)dy = 0$. Then

- (A) e^x is an integrating factor
- (B) e^{-x} is an integrating factor
- (C) $3x$ is an integrating factor
- (D) x^3 is an integrating factor

Question 13 (JAM 2008 • Q14 • ID: JAM_2008_Q14)

MCQ

+6 M

Neg: -2.0

Q.14 Let $y_1(x)$ and $y_2(x)$ be twice differentiable functions on an interval I satisfying the differential equations $\frac{dy_1}{dx} - y_1 - y_2 = e^x$ and $2\frac{dy_1}{dx} + \frac{dy_2}{dx} - 6y_1 = 0$. Then $y_1(x)$ is

(A) $C_1 e^{-2x} + C_2 e^{3x} - \frac{1}{4} e^x$

(B) $C_1 e^{2x} + C_2 e^{-3x} + \frac{1}{4} e^x$

(C) $C_1 e^{2x} + C_2 e^{-3x} - \frac{1}{4} e^x$

(D) $C_1 e^{-2x} + C_2 e^{3x} + \frac{1}{4} e^x$

Question 14 (JAM 2008 • Q9 • ID: JAM_2008_Q9)

MCQ

+6 M

Neg: -2.0

Q.9 Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y'' + P(x)y' + Q(x)y = 0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I . Then $y_3(x) = a y_1(x) + b y_2(x)$ and $y_4(x) = c y_1(x) + d y_2(x)$ are linearly independent solutions of the given differential equation if

(A) $ad = bc$

(B) $ac = bd$

(C) $ad \neq bc$

(D) $ac \neq bd$

Question 15 (JAM 2007 • Q14 • ID: JAM_2007_Q14)

MCQ

+6 M

Neg: -2.0

14. One of the integrating factors of the differential equation $(y^2 - 3xy)dx + (x^2 - xy)dy = 0$ is

(A) $1/(x^2 y^2)$

(B) $1/(x^2 y)$

(C) $1/(x y^2)$

(D) $1/(x y)$

Question 16 (JAM 2007 • Q2 • ID: JAM_2007_Q2)

MCQ

+6 M

Neg: -2.0

2. If k is a constant such that $xy + k = e^{(x-1)^2/2}$ satisfies the differential equation

$$x \frac{dy}{dx} = (x^2 - x - 1)y + (x - 1), \text{ then } k \text{ is equal to}$$

(A) 1

(B) 0

(C) -1

(D) -2

Question 17 (JAM 2006 • Q12 • ID: JAM_2006_Q12)

MCQ

+6 M

Neg: -2.0

12. The differential equation representing the family of circles touching y-axis at the origin is

- (A) linear and of first order
- (B) linear and of second order
- (C) nonlinear and of first order
- (D) nonlinear and of second order

Question 18 (JAM 2006 • Q10 • ID: JAM_2006_Q10)

MCQ

+6 M

Neg: -2.0

10. If $(c_1 + c_2 \ln x)/x$ is the general solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + kx \frac{dy}{dx} + y = 0, \quad x > 0,$$

then k equals

- (A) 3
- (B) -3
- (C) 2
- (D) -1

Question 19 (JAM 2005 • Q3 • ID: JAM_2005_Q3)

MCQ

+6 M

Neg: -2.0

3. The general solution of $x^2 \frac{d^2 y}{dx^2} - 5x \frac{dy}{dx} + 9y = 0$ is

- (A) $(c_1 + c_2 x)e^{3x}$
- (B) $(c_1 + c_2 \ln x)x^3$
- (C) $(c_1 + c_2 x)x^3$
- (D) $(c_1 + c_2 \ln x)e^{x^3}$

(Here c_1 and c_2 are arbitrary constants.)

Question 20 (JAM 2005 • Q2 • ID: JAM_2005_Q2)

MCQ

+6 M

Neg: -2.0

2. An integrating factor of $x \frac{dy}{dx} + (3x + 1)y = xe^{-2x}$ is

(A) xe^{3x}

(B) $3xe^x$

(C) xe^x

(D) x^3e^x

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	D
2	MCQ	+2	B
3	MCQ	+2	D
4	MCQ	+1	B
5	MCQ	+6	C
6	MCQ	+6	B
7	MCQ	+6	A
8	MCQ	+6	B
9	MCQ	+6	D
10	MCQ	+6	B

Q. No.	Type	Marks	Official Key
11	MCQ	+6	A
12	MCQ	+6	D
13	MCQ	+6	C
14	MCQ	+6	C
15	MCQ	+6	B
16	MCQ	+6	A
17	MCQ	+6	C
18	MCQ	+6	A
19	MCQ	+6	B
20	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.